Incision and drainage for cutaneous abscess/infected sebaceous cyst under local anaesthetic

Simon Jeffs

Advanced Practitioner in general surgery

- Ambulatory and same day emergency care are evolving to meet the increased demand and reduce inpatient admissions.

- Within the local trust ACPs have taken over the management of the surgical emergency clinic which will triage, assess and manage patients who have been referred by either their GP or ED for specific pathway lead treatment such as painless jaundice, gallstone disease and abscess management.

- Cutaneous skin abscesses are a common presentation to both the Emergency Department (ED) and to primary care settings. Statistical trends indicate a growing incidence (Dunn *et al.,* 2021).

- An uncomplicated abscess is caused by a collection of pus within the skin's structures, which is bacterial in origin. The majority of diagnoses are made through a thorough history and examination of the skin and surrounding structures (Thomas *et al.,* 2020)

- The gold standard treatment and management of cutaneous abscesses is incision and drainage (I&D) of the affected area (Stevens *et al.*, 2014).

- In the United Kingdom (UK), there are no specific best practice guidelines for the management of cutaneous abscesses (Elkadi *et al.*, 2023), although the National Institute of Clinical Excellence (NICE) (2022) recommend all patients with an abscess should attend a same-day surgical unit for I&D.

- Following the COVID-19 pandemic, there has been a shift in practice and more abscesses are managed under local anaesthetic (LA) (Elkadi *et al.*, 2023).

- Patient choice is paramount

- I&D is a competency that can be undertaken by Advanced Practitioners (APs) once training has been provided and a safe level of competence achieved (Nicka and Swanson-Biearman, 2019). Htay and Whitehead (2021) acknowledge that APs taking on an extended scope of practice, can provide a service that is cost effective, efficient, and enhances quality of care whilst improving patient overall satisfaction.

- This enables surgeons to focus on providing a higher level of care to critically ill patients without added distraction and pressure within the acute setting (Richards *et al.,* 2018).

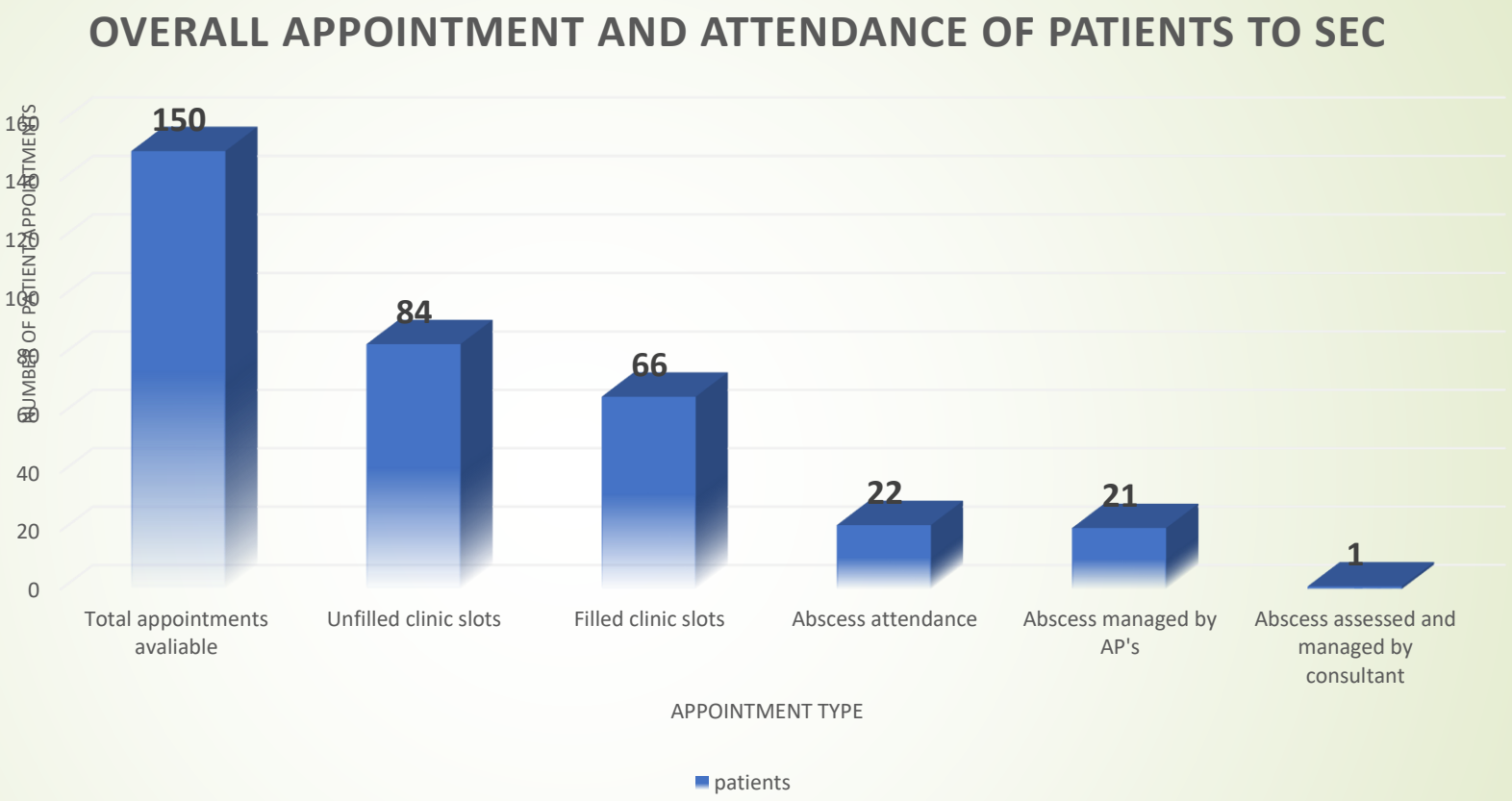
- We have a responsibility to work within their scope of practice at all times (nursing and midwifery council (NMC), 2018; Health and Care Professions Council (HCPC), 2023).

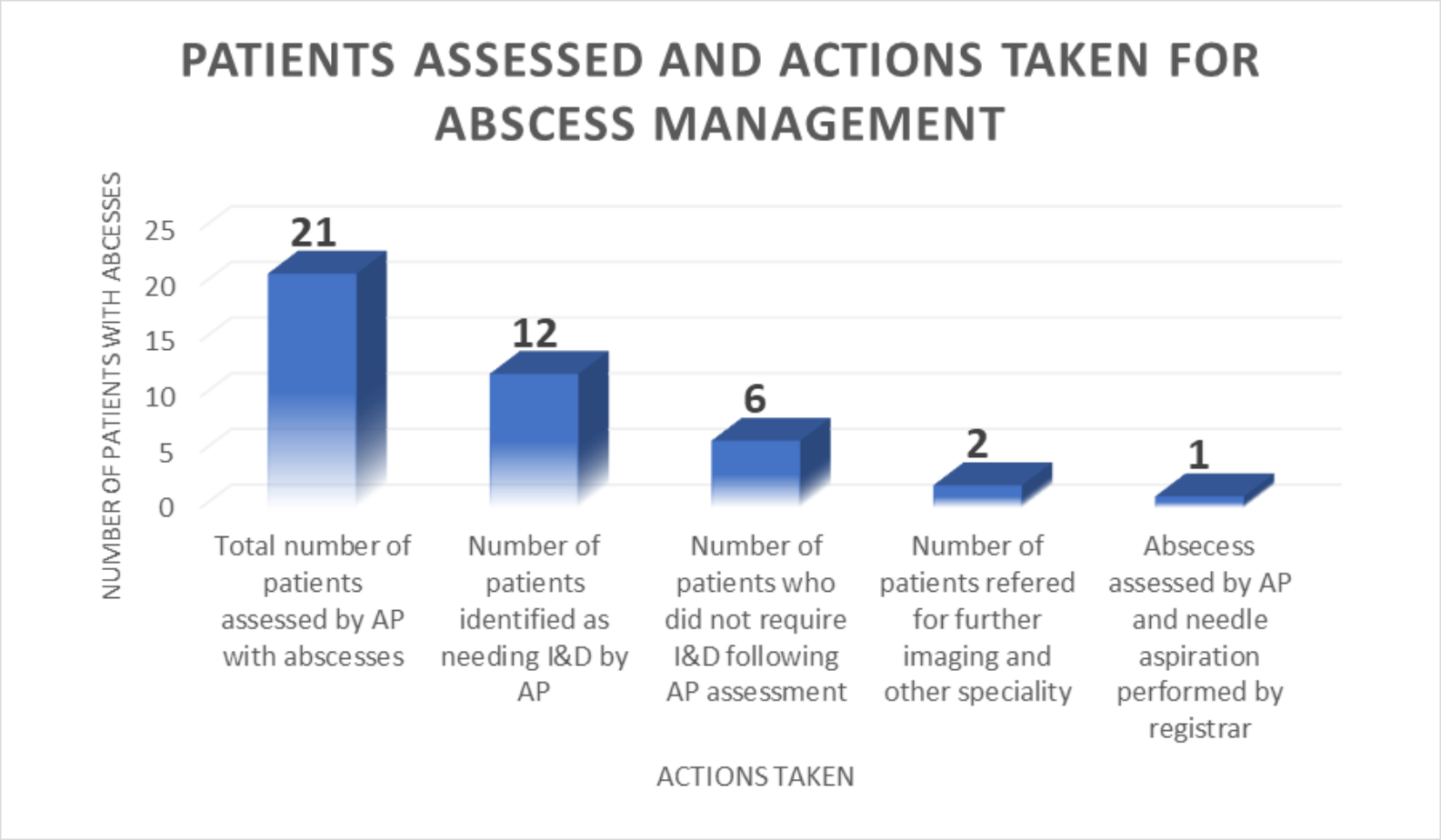
- At present, within the local Trust, there is no guidance to support incision and drainage being undertaken by non-surgeons. Currently there is no governance structure nor competency to support APs to undertake this extended skill. The creation of a standard operating procedure (SOP) forms part of a risk assessment, reducing the risk of 'never events' from occurring, and provides boundaries and standardisation on how a to perform the procedure (Morgan *et al.*, 2015).

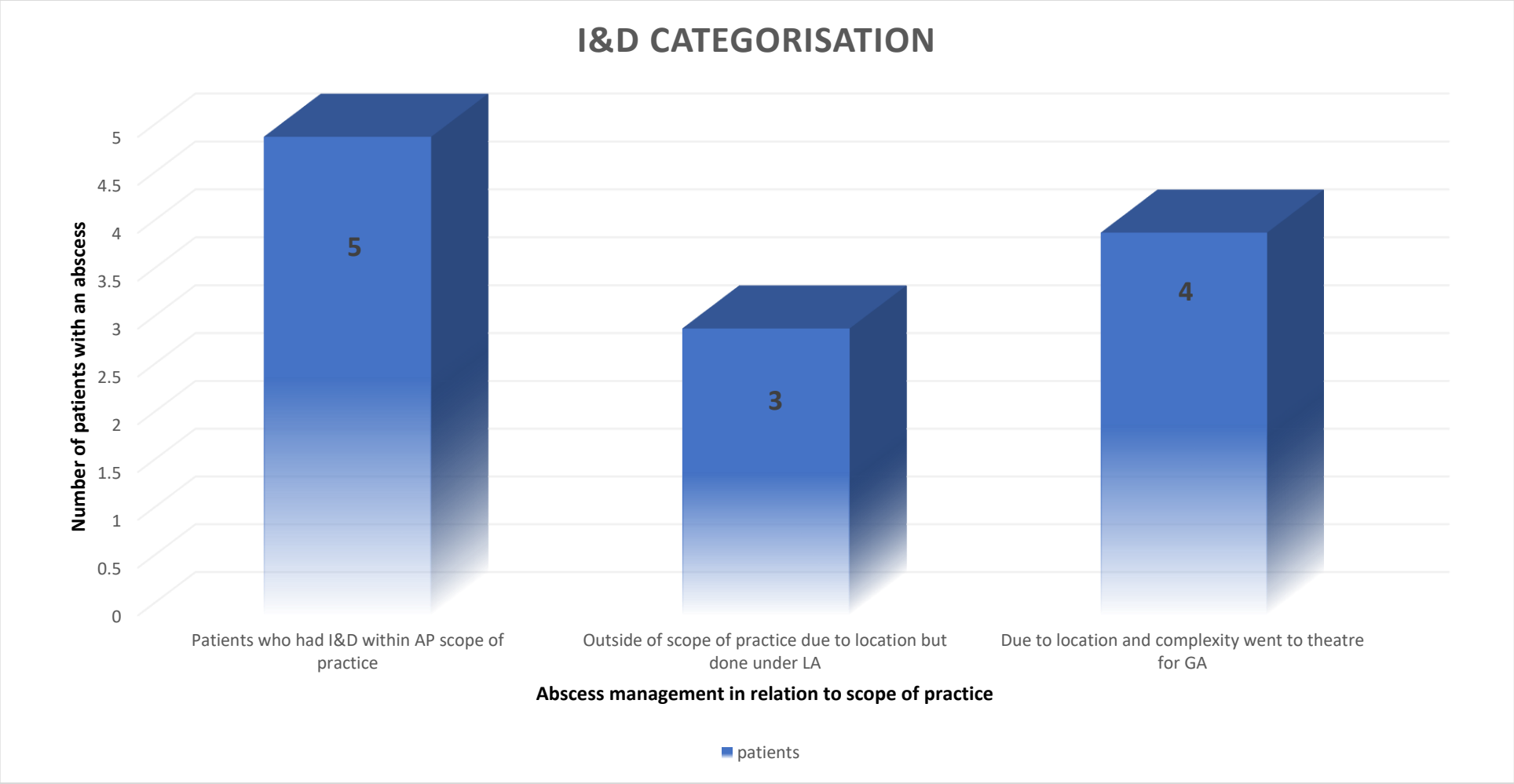
Audit template

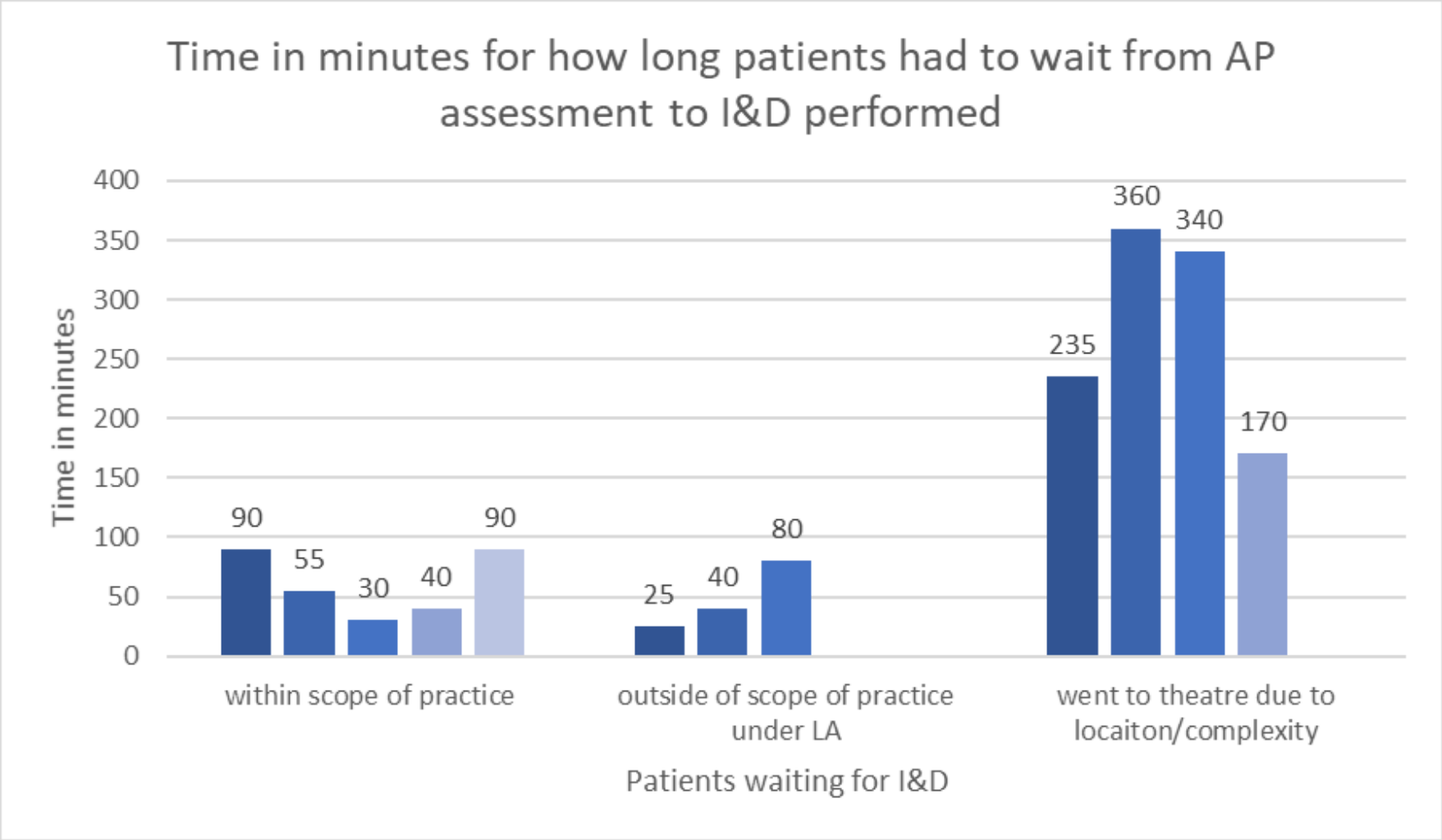
Information gathering for incision and drainage by AP's and PA's

Location of abscess/infected sebaceous cyst	Time of assessment by AP	Time of I&D procedure performed by trained individual	was I&D necessary Yes/No If No, document management plan	I&D performed under Local or GA	Time between referral and assessment	Length of time on ABX i.e. initial GP assessment

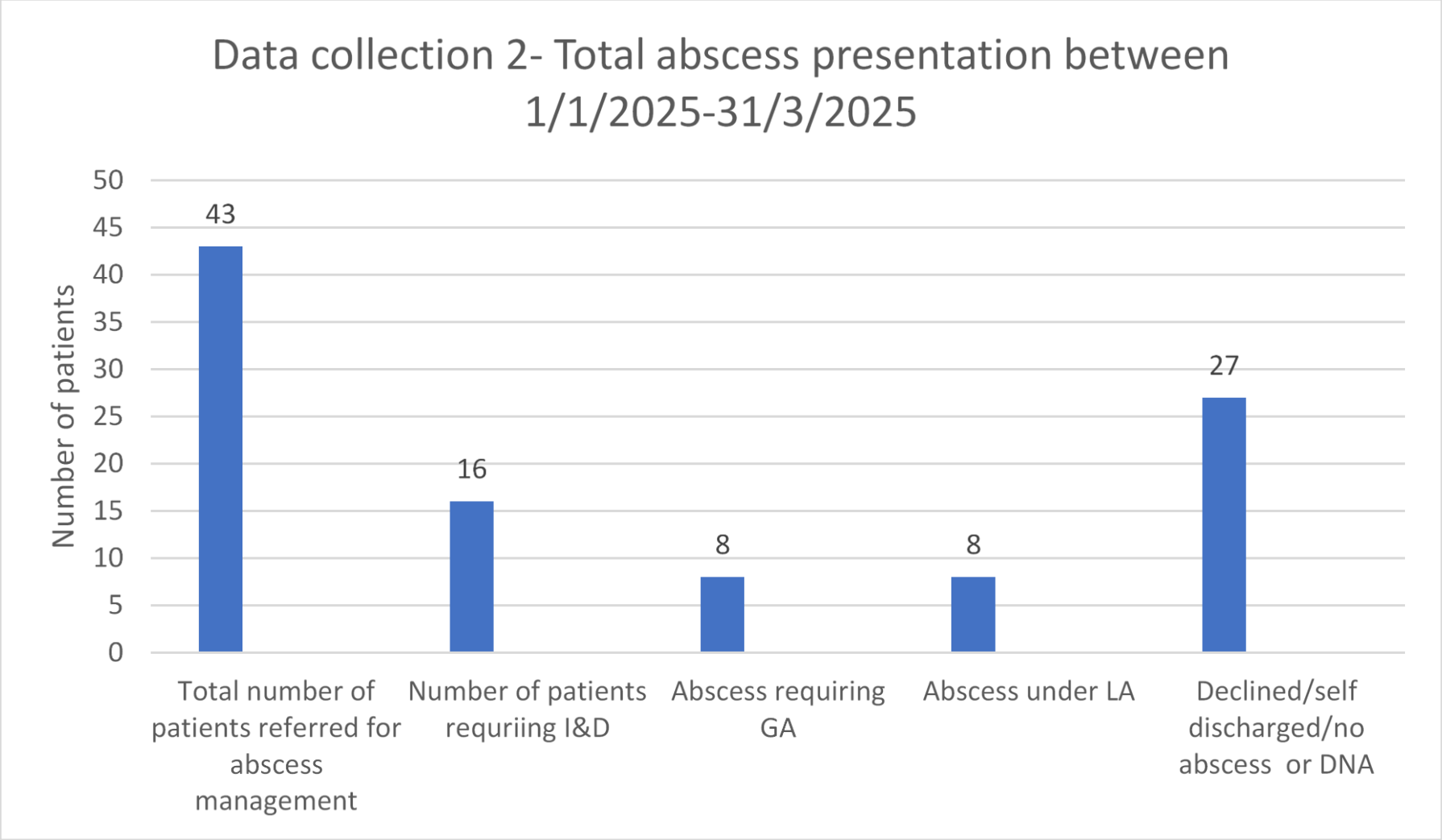

OVERALL APPOINTMENT AND ATTENDANCE OF PATIENTS TO SEC
NUMBER OF PATIENTS APPOINTMENTS
0
20
40
60
80
100
120
140
160
150
84
66
22
21
1
Total appointments avaliable
Unfilled clinic slots
Filled clinic slots
Abscess attendance
Abscess managed by AP's
Abscess assessed and managed by consultant
APPOINTMENT TYPE
patients

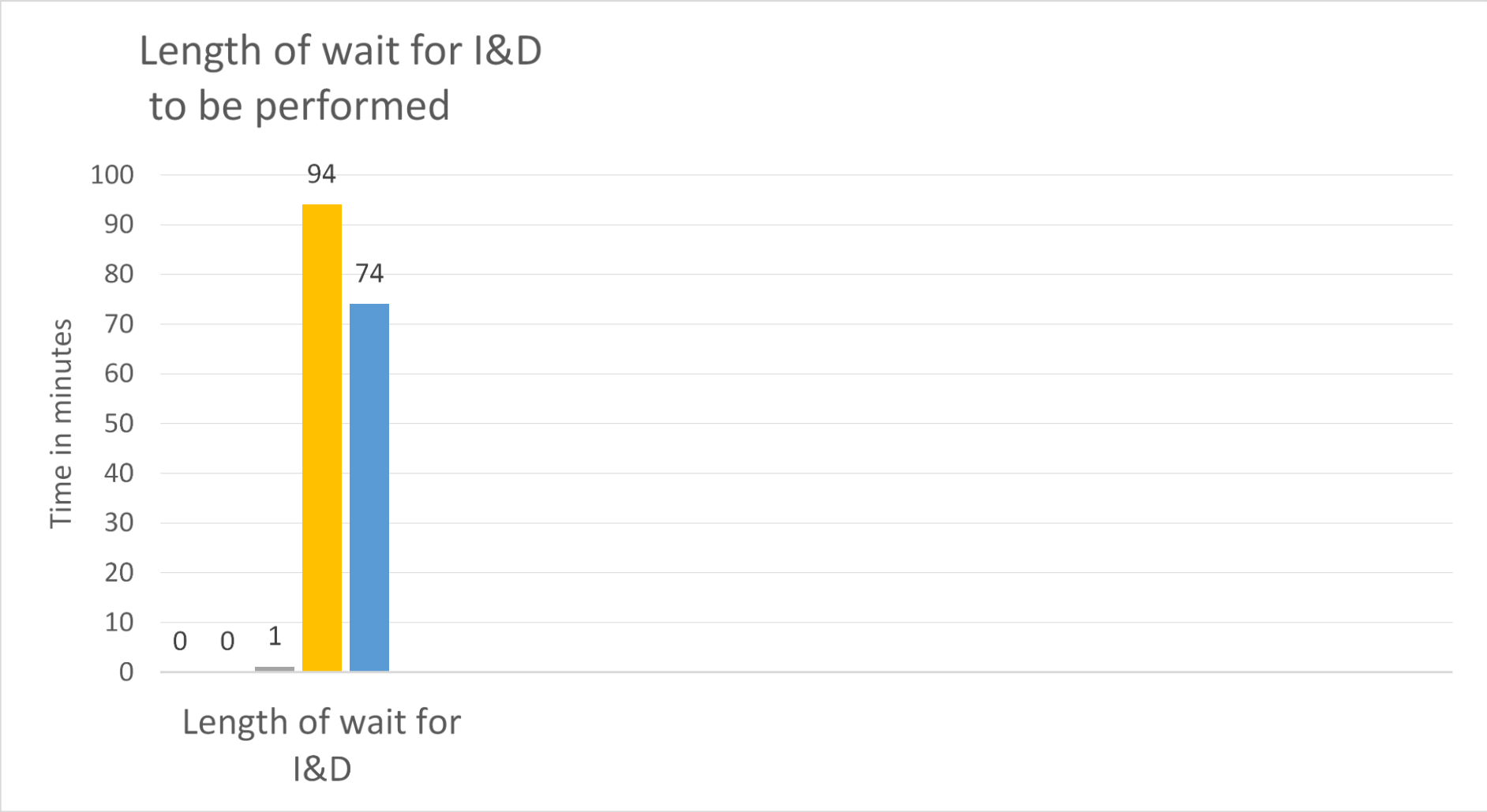

PATIENTS ASSESSED AND ACTIONS TAKEN FOR ABSCESS MANAGEMENT
NUMBER OF PATIENTS WITH ABCESSES
25
20
15
10
5
0
21
12
6
2
1
Total number of patients assessed by AP with abscesses
Number of patients identified as needing I&D by AP
Number of patients who did not require I&D following AP assessment
Number of patients refered for further imaging and other speciality
Absecess assessed by AP and needle aspiration performed by registrar
ACTIONS TAKEN


I&D CATEGORISATION
Number of patients with an abscess
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
5
3
4
Patients who had I&D within AP scope of practice
Outside of scope of practice due to location but done under LA
Due to location and complexity went to theatre for GA
Abscess management in relation to scope of practice
patients


Time in minutes for how long patients had to wait from AP assessment to I&D performed
Time in minutes
400
350
300
250
200
150
100
50
0
90
55
30
40
90
25
40
80
235
360
340
170
within scope of practice
outside of scope of practice under LA
went to theatre due to locaiton/complexity
Patients waiting for I&D

	Date	Location of abscess	Time of assessment	Time of I&D completion	Reason for delay	Performed under local or GA	Any other comments
1							


Data collection 2- Total abscess presentation between 1/1/2025-31/3/2025
Number of patients
50
45
40
35
30
25
20
15
10
5
0
43
16
8
8
27
Total number of patients referred for abscess management
Number of patients requriing I&D
Abscess requiring GA
Abscess under LA
Declined/self discharged/no abscess or DNA


Length of wait for I&D to be performed
Time in minutes
100
90
80
70
60
50
40
30
20
10
0
0
0
1
94
74
Length of wait for I&D

What next

- To submit audit report to Clinical Audit & Effectiveness Team for review.
- To present findings at surgical governance
- Undertake official training
- Get signed off as competent

Criteria	Knowledge and understanding	Achieved
Indication/ Informed Consent:	Understands the general indications, contraindications, potential complications, and clinical value of procedure; able to explain to patient	☐
Knowledge of Specific Procedure	Demonstrates familiarity with all aspects of procedure	☐
Procedure Setup	Gathers key instruments and supplies; selects appropriate size needle/syringe; properly positions patient; identifies landmarks and needle entry site; maintains sterile technique	☐
Local Anesthesia	Able to identify risk factors and contraindication to lidocaine. Ensures local anaesthetic doses are within maximum permitted doses, based on patient weight. Can efficiently perform a field block without guidance and with good anaesthesia. Tests that the local has worked before proceeding	☐ ☐ ☐ ☐
Flow of Procedure	Demonstrates forward planning with reasonable progression of procedure	☐
Incision & Respect for Tissue	Appropriate incision size and depth, careful handling of tissue using appropriate force; clears all drainable pockets and irrigates with saline	☐
Packing Placement	Adequate packing of wound without guidance and covers with a suitable sterile dressing	☐
Post procedure	Correctly disposes of waste Provides patient instructions for on-going review in the community and/or review in AAA. Give advice on simple analgesia and identify if antibiotics are needed	☐ ☐ ☐
Management of Complications	Recognizes and appropriately addresses developing complication; halts the procedure appropriately.	☐

References:

- Dunn, M., Savoie, K, Erdem, G., Dykes, M.W., Buckingham, D., Spencer, S. Besner, G. and Kenney, G. (2021) 'Quality improvement methodology can reduce hospitalisation for abscess management', *Emergency Medical Journal*, 39, pp. 788-794. Available at: https://emj.bmj.com/content/emermed/39/10/788.full.pdf
- Elkadi, H.H., Dodd, E., Poulton, T., Bolton, W., Burke, J.R., Tiernan, J. and Jayne, D. (2023) 'The effect of anaesthetic use on healing in subcutaneous abscess management: A retrospective before and after cohort study', *Annals Royal College of surgeons England*, 105, pp. 241-246. Available at: https://publishing.rcseng.ac.uk/doi/full/10.1308/rcsann.2021.0362
- Health and Care Professions Council (HCPC) (2023) *The Standards of Proficiency for Paramedics.* Available at: https://www.hcpc-uk.org/standards/standards-of-proficiency/paramedics/
- Htay, M. and Whitehead, D. (2021) 'The effectiveness of the role of advanced nurse practitioners compared to physician-led or usual care: A systematic review', *International Journal of Nursing Studies Advances*, 3, 100034. Available at: https://wwwsciencedirectcom.plymouth.idm.oclc.org/science/article/pii/S2666142X21000163
- Morgan, L., New, S., Robertson, E., Collins, G., Rivero-Arias, O., Catchpole, K., Pickering, S.P., Hadi, M., Griffin, D. and McCulloch, P. (2015) 'Effectiveness of facilitated introduction of a standard operating procedure into routine processes in the operating theatre: A controlled interrupted time series', *BMJ Quality Safety*, 24, pp. 120-127. Available at: https://qualitysafety.bmj.com/content/qhc/24/2/120.full.pdf
- National Institute of Clinical Excellence (NICE) (2022) *Boils, Carbuncles, and Staphylococcal Carriage*. Available at: https://cks.nice.org.uk/topics/boils-carbuncles-staphylococcal-carriage/
- Nicka, A. and Swanson-Biearman, B. (2019) 'Development of a subcutaneous abscess simulator for incision and drainage', *Clinical Simulation in Nursing*, 37, pp. 40-43. Available at: https://wwwsciencedirectcom.plymouth.idm.oclc.org/science/article/pii/S1876139919300684
- nursing and midwifery council (NMC) (2018) 'The Code'. Available at: https://www.nmc.org.uk/standards/code/
- Richards, S., Thompson, D., Holmes, M., Mitchell, A. and Mr Saha, A. (2018) *Surgical Ambulatory Emergency Care Network Toolkit*. Available at: https://www.ambulatoryemergencycare.org.uk/uploads/files/1/News/SAEC-TOOLKIT-version-1---September-2018.pdf
- Stevens, D.L, Bisno, A.L., Chambers, H.F., Dellinger, E.P., Goldstein, E.J.C., Gorbach, S.L., Hirschmann, J.V., Kaplan, S.A., Montoya, J.G. and Wade, J.C. (2014) 'Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America', *Clinical Infectious disease*, 59, pp. e10-e52. Available at: https://academic.oup.com/cid/article/59/2/147/442347?login=false
- Thomas, O., Ramsay, A., Yiasemidou, M., Hardie, C., Ashmore, D., Macklin, C., Bandyopadhyay, D., Patel, B., Burke, J.R. and Jayne, D. (2020) 'The surgical management of cutaneous abscesses: A UK cross-sectional survey', *Annals of Medicine and Surgery*, 60, pp. 654–659. Available at: https://www.clinicalkey.com/#!/content/playContent/1-s2.0-S2049080120304957

ANY
QUESTIONS?
WHEN?
WHO?
WHAT?
WHERE?
When?
Where?
What?
HOW?
Why?